🔬 General Science — Master File (Physics · Chemistry · Biology) · SSC CGL

> **The single consolidated Science resource** for SSC CGL — the *hidden gold mine* (Class 9–10 level, high-scoring). Physics, Chemistry, Biology + a Space/Tech touch, with PYQ question-types, curated & new questions. 🔴 High per the PYQ Blueprint.

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1. How SSC Asks Science (PYQ Types)

PYQs are the blueprint — the concepts stay identical year to year, only the wording changes. The recurring question shapes:

Type	Example stem	Prep
SI unit match	"SI unit of force is —"	Units table (§2)
Disease → cause	"Malaria is caused by —"	Viral/bacterial/protozoan lists (§4)
Vitamin → deficiency	"Scurvy is due to lack of —"	Vitamin table (§4)
Which element/allotrope	"Only liquid metal —"	Element facts (§3)
Everyday phenomenon	"Sky is blue because —"	Optics/scattering (§2)
Body facts	"Powerhouse of the cell —"	Human body (§4)

2. Physics

Motion & Force

- **Newton's Laws:** 1st (inertia) · 2nd (**F = ma**, force = rate of change of momentum) · 3rd (action–reaction).
- Speed = distance/time (scalar) · Velocity = displacement/time (vector) · Momentum **p = mv**.
- Work **W = F·d** · Power = Work/time · **g ≈ 9.8 m/s²** · Escape velocity of Earth **≈ 11.2 km/s**.

Light & Optics

- Speed of light in vacuum **≈ 3 × 10⁸ m/s**; travels in straight lines.

- **Concave mirror** converges · **Convex mirror** diverges (**vehicle rear-view** — wider field).
- **Convex lens** converges (corrects **hypermetropia**/long-sight) · **Concave lens** diverges (corrects **myopia**/short-sight).
- **Dispersion:** white light → **VIBGYOR** via prism (Red bends least, Violet most).
- **Sky is blue** → Rayleigh **scattering**; sunsets red (red scatters least).

Sound

- **Longitudinal mechanical wave**; needs a medium (no sound in vacuum).
- Speed in air ≈ 343 m/s; **faster in solids > liquids > gases**.
- Audible range **20 Hz – 20 kHz**; below = infrasonic, above = **ultrasonic** (used in SONAR).

Electricity & SI Units

- **Ohm's Law: V = IR** · Power P = VI = I^2R · **1 unit = 1 kWh**.
- AC for transmission; DC from batteries; **fuse** melts to protect circuits.

Quantity	SI unit
Force	Newton (N)
Energy/Work	Joule (J)
Power	Watt (W)
Pressure	Pascal (Pa)
Frequency	Hertz (Hz)
Current	Ampere (A)
Resistance	Ohm (Ω)
Charge	Coulomb (C)
Temperature	Kelvin (K)
Magnetic flux density	Tesla (T)

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3. Chemistry

Acids, Bases & Salts

- **Acids:** sour, turn blue litmus red, **pH < 7** (HCl, H_2SO_4, citric acid).
- **Bases/Alkalis:** bitter, slippery, turn red litmus blue, **pH > 7** (NaOH, NH_4OH).
- **pH scale 0–14**; pure water neutral (**pH 7**). Acid + base → salt + water (neutralisation).
- Sulphuric acid = "King of Chemicals" · Baking soda = **$NaHCO_3$** · Washing soda = **Na_2CO_3**.

Elements & Periodic Table

- **Mendeleev** — periodic table by atomic mass; **Moseley** — modern (by atomic number).
- Atom: protons (+) & neutrons in nucleus, electrons (−) in shells. **Atomic number = protons.**
- Most abundant: **Earth's crust → Oxygen**; **universe → Hydrogen**.
- Lightest element **Hydrogen** · lightest metal **Lithium** · heaviest natural **Uranium**.
- **Mercury (Hg)** — only metal liquid at room temp · **Bromine** — only liquid non-metal.
- **Diamond & Graphite** — allotropes of **carbon** (diamond = hardest natural substance).
- Formulas: Water H_2O · Salt NaCl · CO_2 · Ozone O_3 · Quicklime CaO · Slaked lime $Ca(OH)_2$.

4. Biology

Human Body

- **Cell** = basic unit of life (Robert Hooke coined "cell"). **Mitochondria** = "powerhouse"; **Ribosomes** = protein synthesis. **DNA** = Deoxyribonucleic Acid (genetic info).
- Largest organ **Skin** · largest internal organ/gland **Liver** · smallest bone **Stapes** (ear) · longest bone **Femur** · strongest muscle **Masseter** (jaw).
- Adult bones **206** (baby ~300). Blood = plasma + RBCs (O_2 via haemoglobin) + WBCs (immunity) + platelets (clotting). **RBC/WBC/platelets made in bone marrow.** Blood pH ≈ **7.4**.
- **Universal donor O−** · **universal recipient AB+** · body temp **37 °C / 98.6 °F**.
- Heart = 4 chambers, pulse ~72/min · Kidneys filter blood · **Insulin** (pancreas) regulates blood sugar · digestion begins in the **mouth** (salivary amylase).

Vitamins & Deficiency Diseases

Vitamin	Deficiency disease
A (Retinol)	Night blindness
B1 (Thiamine)	Beriberi
B3	Pellagra
B12	Anaemia
C (Ascorbic acid)	**Scurvy**
D (Calciferol, "sunshine")	**Rickets** / Osteomalacia
K	Poor blood clotting

Fat-soluble: **A, D, E, K** · Water-soluble: **B, C**.

Diseases by Cause

- **Viral:** AIDS, Influenza, Dengue, Chickenpox, Polio, Rabies, COVID-19, Measles, Mumps.
- **Bacterial:** TB, Cholera, Typhoid, Tetanus, Leprosy, Plague.
- **Protozoan:** Malaria (**Plasmodium**, female Anopheles), Kala-azar.
- **Deficiency:** Goitre (iodine), Anaemia (iron).

Nutrition

- Macronutrients: Carbohydrates (energy), **Proteins (growth/repair, made of amino acids)**, Fats (energy store).
- **Photosynthesis:** chlorophyll + sunlight + CO_2 + water → glucose + O_2.

5. Space & Technology (quick)

- **ISRO** founded 1969; HQ Bengaluru; first satellite **Aryabhata (1975)**.
- **Chandrayaan-3 (2023)** — India first to soft-land near the lunar **south pole**. **Mangalyaan (MOM, 2013)** — Mars, first attempt success.
- Launch vehicles: **PSLV** (polar), **GSLV** (geostationary, cryogenic). Satellite series: INSAT, IRS, NAVIC (regional navigation).

- Newton, Einstein (relativity, $E=mc^2$), C.V. Raman (Raman Effect, Nobel 1930), Homi Bhabha (father of Indian nuclear programme), Vikram Sarabhai (father of Indian space programme).

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6. Curated PYQ-Pattern Questions

Q1. SI unit of force — (a) Joule (b) Watt (c) **Newton** (d) Pascal

Q2. Ohm's Law — (a) V=I/R (b) **V=IR** (c) I=VR (d) R=VI

Q3. Bending of light between media — (a) reflection (b) **refraction** (c) dispersion (d) diffraction

Q4. Sky appears blue due to — (a) reflection (b) refraction (c) **scattering** (d) absorption

Q5. Only metal liquid at room temp — (a) Bromine (b) **Mercury** (c) Sodium (d) Lithium

Q6. pH of a neutral solution — (a) 0 (b) **7** (c) 10 (d) 14

Q7. Gas that turns lime water milky — (a) O_2 (b) H_2 (c) **CO_2** (d) N_2

Q8. Diamond & graphite are allotropes of — (a) Sulphur (b) **Carbon** (c) Phosphorus (d) Silicon

Q9. Powerhouse of the cell — (a) Nucleus (b) Ribosome (c) **Mitochondria** (d) Golgi

Q10. Bones in an adult body — (a) 196 (b) **206** (c) 216 (d) 300

Q11. Deficiency of Vitamin C causes — (a) Rickets (b) **Scurvy** (c) Night blindness (d) Beriberi

Q12. "Sunshine vitamin" — (a) A (b) C (c) **D** (d) K

Q13. Malaria is caused by — (a) Bacteria (b) Virus (c) **Plasmodium** (d) Fungus

Q14. Universal donor blood group — (a) AB+ (b) **O−** (c) A+ (d) B−

Q15. Hormone regulating blood sugar — (a) Adrenaline (b) Thyroxine (c) **Insulin** (d) Oestrogen

Q16. 1 unit of electrical energy — (a) Watt (b) Volt (c) **kWh** (d) Ampere

Q17. Bacterial disease — (a) Dengue (b) **Tuberculosis** (c) AIDS (d) Influenza

Q18. Most abundant element in Earth's crust — (a) Silicon (b) **Oxygen** (c) Aluminium (d) Iron

Q19. Goitre is caused by deficiency of — (a) Iron (b) **Iodine** (c) Calcium (d) Vitamin D

Q20. Mirror used as vehicle rear-view — (a) Concave (b) **Convex** (c) Plane (d) Cylindrical

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7. Freshly-Created Practice Questions

New questions in the SSC pattern; answer key at the end.

N1. The escape velocity of Earth is approximately —

(a) 9.8 km/s (b) 11.2 km/s (c) 7.9 km/s (d) 3×10^8 m/s

N2. A concave lens is used to correct —

(a) hypermetropia (b) myopia (c) astigmatism (d) cataract

N3. Ultrasonic sound has a frequency —

(a) below 20 Hz (b) 20 Hz–20 kHz (c) above 20 kHz (d) exactly 20 kHz

N4. The most abundant element in the universe is —

(a) Oxygen (b) Hydrogen (c) Helium (d) Carbon

N5. Washing soda is chemically —

(a) $NaHCO_3$ (b) Na_2CO_3 (c) NaOH (d) NaCl

N6. Red blood cells are produced in the —

(a) liver (b) kidney (c) bone marrow (d) heart

N7. The longest bone in the human body is the —

(a) Stapes (b) Femur (c) Tibia (d) Humerus

N8. Beriberi is caused by the deficiency of vitamin —

(a) A (b) B1 (c) C (d) D

N9. Which of the following is a protozoan disease?

(a) Tuberculosis (b) Dengue (c) Kala-azar (d) Tetanus

N10. India's first satellite, launched in 1975, was —

(a) Bhaskara (b) Aryabhata (c) Rohini (d) INSAT-1A

Answer Key (Section 7):

N1-(b) 11.2 km/s · N2-(b) myopia · N3-(c) above 20 kHz · N4-(b) Hydrogen · N5-(b) Na_2CO_3 · N6-(c) bone marrow · N7-(b) Femur · N8-(b) B1 · N9-(c) Kala-azar · N10-(b) Aryabhata

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8. Rapid Revision Sheet

- $F = ma$ · $p = mv$ · $W = F \cdot d$ · $g \approx 9.8\ m/s^2$ · escape velocity 11.2 km/s · light 3×10^8 m/s
- Convex lens → hypermetropia · Concave lens → myopia · Convex mirror → rear-view (wide) · Blue sky → scattering
- Ohm's Law V=IR · 1 unit = kWh · SI: Force-N, Energy-J, Power-W, Pressure-Pa, Frequency-Hz
- pH 7 neutral; acids <7, bases >7 · baking soda $NaHCO_3$, washing soda Na_2CO_3
- Mercury → only liquid metal · Bromine → only liquid non-metal · Diamond/Graphite → carbon allotropes
- Abundant: Earth's crust → Oxygen; universe → Hydrogen
- Mitochondria → powerhouse · 206 bones · Stapes smallest, Femur longest · RBC in bone marrow · blood pH 7.4
- Universal donor O−, recipient AB+ · body temp 37 °C · Insulin → blood sugar (pancreas)
- Scurvy → C · Rickets → D · Night blindness → A · Beriberi → B1 · Goitre → iodine · Anaemia → iron
- Malaria → Plasmodium (Anopheles) · Viral: AIDS/Dengue/Flu · Bacterial: TB/Cholera/Typhoid
- ISRO 1969, Aryabhata 1975 · Chandrayaan-3 (2023, lunar south pole) · PSLV/GSLV · C.V. Raman (Raman Effect)

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